2020a; Wulfe et al., 2022). However, in most cases, reward modeling is used only when the true reward function is not known, making direct evaluation impossible. Hence, the reward model is typically evaluated in an *indirect* way by optimizing an RL policy using the learned reward model and then evaluating the generations from the RL policy. This makes the reward model evaluation intricately dependent on the policy optimization process which is inherently expensive and noisy. It is also not clear how robust a reward model evaluation is to many ad-hoc choices made in the policy optimization process: e.g., choice of RL algorithm, policy network architecture, compute spent, and other various hyperparameter choices (Gao et al., 2022). Another issue with indirect evaluation is that the evaluation signal for the reward model is the same as the training signal – human approval. As a result, training and evaluation failures will be correlated. Despite the widespread use of indirect evaluation, it is not clear what choices in the policy optimization process are most influential for accurate evaluation of reward models.

3.3 Challenges with the Policy

Here, we discuss challenges from policy optimization, misgeneralization, power-seeking, and mode collapse. Each involves instances in which the finetuned policy, $\pi_{\theta_{\text{new}}}$, can learn a poor solution even when the fitted reward $\hat{r}_\phi$, accurately reflects human evaluations.

3.3.1 Robust Reinforcement Learning is Difficult

Safety in deployment requires robust performance, yet it remains challenging simply to train AI systems using RL.

Tractable: It is (still) challenging to optimize policies effectively. RL agents must interact with the environment to collect their own data. This requires balancing exploratory and exploitatory behavior (Amin et al., 2021; Yang et al., 2021). Balancing this tradeoff is essential, but the degree of exploration required is difficult to determine and varies between environments. This is further complicated in settings with high-dimensional state/action spaces or sparse rewards (Ding and Dong, 2020). Balancing exploration and exploitation in deep RL remains a fundamental yet open challenge (Amin et al., 2021; Yang et al., 2021). Deep RL is unstable, and results are often highly sensitive to initialization and difficult to reproduce (Nikishin et al., 2018; Irpan, 2018; Henderson et al., 2018). This instability is attributed to multiple factors such as the random nature of exploration, the violation of the i.i.d assumption in data collection, the biased nature of value functions, and the general unpredictability of learning in deep neural networks (Amin et al., 2021). Uc-Cetina et al. (2023) overview methods and limitations for RL with LLMs in particular.

Tractable: Policies tend to be adversarially exploitable. Even when learned policies are trained with a perfect reward signal, perform well at the task they are trained for, and generalize to a wide range of scenarios, they can still perform poorly in adversarial situations. This is a pressing concern, as models deployed into the real world can be adversarially attacked by humans or other AI systems. Even "superhuman" policies can fail catastrophically against policies specifically designed to exploit them (Gleave et al., 2020b; Wu et al., 2021b; Wang et al., 2022). Adversarial policies can be found either by re-purposing existing deep-reinforcement learning algorithms or by manual human optimization in the case of prompt-injections and jailbreaks (Willison, 2023; Albert, 2023; Oneal, 2023; Li et al., 2023a; Wolf et al., 2023; Liu et al., 2023; Rao et al., 2023; Wei et al., 2023; Shen et al., 2023) for language-models. Black-box access to a model (e.g., via API access) is sufficient for many adversarial policy attack algorithms, though white-box access (enabled for example by open-sourced or leaked model weights) enables even stronger exploits (Kos and Song, 2017; Casper et al., 2022).

3.3.2 Policy Misgeneralization

Fundamental: Policies can perform poorly in deployment even if rewards seen during training were perfectly correct. The deployment distribution can always differ from the training and evaluation distributions in real-world settings (Christiano, 2019). Even with a correct reward signal, a policy can learn to competently pursue the wrong goal whenever the true goal is correlated with other events. Shah et al. (2022); Di Langosco et al. (2022) and Hilton et al. (2020) study this type of failure in-depth. Shah et al.

(2022) present an example scenario in which a systems trained with RLHF misgeneralizes to pursue the mechanism of reward administration itself instead of the intended goal.

Fundamental: Optimal RL agents tend to seek power. RL agents have an incentive to seek power when possible to help them accomplish their goals (Turner, 2021; Turner et al., 2019; Turner and Tadepalli, 2022; Ngo, 2022; Krakovna and Kramar, 2023; Ngo, 2022) Versions of this can emerge from the way that RLHF is typically used to finetune LLMs. For example, a question-answering LLM trained with RLHF would be incentivized to influence human interlocutors in order to avoid conversations about challenging topics. Sycophantic behavior from LLMs offers another example (Perez et al., 2022b).

3.3.3 Distributional Challenges

There are challenges posed by the distribution of outputs produced by the model both before and after training.

Tractable: The pretrained model introduces biases into policy optimization. RLHF in LLMs typically begins with a base model that has been pretrained on internet text. This base model is typically used both as the initialization for the RL policy network and the reference model for KL-regularization. Korbak et al. (2022b) formalizes how RL with these KL penalties can be viewed as a form of Bayesian inference with the base model determining the prior. While empirically useful, it causes the base model to significantly influence the final model. Using a base model that has been pretrained on web text is a convenient initialization – not a principled one. Moreover, internet text encodes harmful biases (e.g., about human demographics), which are then inherited by the downstream model (Weidinger et al., 2021). These biases can persist through RLHF training process. For example, if sounding confident and producing correct answers are correlated in the base model, the reward model will learn that sounding confident is good and reinforce this in the policy.

Tractable: RL contributes to mode collapse. RL finetuning decreases the diversity of samples produced by a model (Khalifa et al., 2021; Perez et al., 2022a; Glaese et al., 2022; Go et al., 2023) (a phenomenon known as "mode collapse"). OpenAI (2023) found that RLHF finetuning of GPT-4 harmed its calibration on question-answering. Santurkar et al. (2023) found LLMs finetuned with RLHF expressed a narrow distribution of political views. Mode collapse is plausibly due in part to switching from the supervised pretraining objective to an RL objective (Song et al., 2023). RL incentivizes the policy to output high-scoring completions with high probability, rather than with a probability in line with a training distribution. Addressing this is complicated because mode collapse can be beneficial or harmful in different cases. For example, it is desirable if an LLM assistant is 90% sure the answer to a question is "yes", it is better for the LLM to answer "probably" 100% of the time rather than answering "yes" 90% of the time and "no" 10% of the time. On the other hand, some preferences are inherently distributional (Khalifa et al., 2021; Weidinger et al., 2021) (e.g., gender balance).

3.4 Challenges with Jointly Training the Reward Model and Policy

RLHF's dependence on training both a reward model and policy poses two unique problems.

Tractable: Joint training induces distribution shifts. Learning both a reward model and a policy is technically challenging – the reward model influences the learned policy, and the policy determines the distribution of the data used to train the reward. On one hand, if the reward model is trained on offline data, it is likely to misgeneralize (Levine et al., 2020). On the other hand, if reward and policy are learned jointly by gathering feedback from policy samples, the system will be prone to "auto-induced distributional shift" (Krueger et al., 2020; Carroll et al., 2022). Features with overestimated rewards will become gradually more present in the feedback data, and features with underestimated rewards will disappear. Thus errors from the reward model can accumulate and become difficult to correct with feedback once the policy stops generating diverse alternatives (Wu et al., 2021a).

Tractable: It is difficult to balance efficiency and avoiding overfitting by the policy. The three key steps of RLHF can be performed synchronously, but in practice with LLMs, they are often performed serially. In this case, the reward model will typically be inaccurate off-distribution, which is precisely where the policy

will learn to go (Gao et al., 2022; Levine et al., 2020). This is usually solved by obtaining fresh preference labels after a certain number of iterations of policy training. Appropriately setting this hyperparameter is important. Too low and information in the preference labels is wasted; too high and the policy navigates to unreliable regions of the reward model (McKinney et al., 2023; Christiano et al., 2017). Without a labeled validation set in the regions the policy is exploring, it is difficult to detect reward over-optimization during training. Helpful approaches might include measuring KL-shift (Gao et al., 2022) or tracking the amount of disagreement in an ensemble of reward models.

4 Incorporating RLHF into a Broader Framework for Safer AI

Because of the challenges surveyed in Section 3, relying heavily on RLHF for developing safe AI poses risks. While RLHF is useful, it does not solve the fundamental challenges of developing human-aligned AI. More generally, no single strategy should be treated as a comprehensive solution. A better approach is defense in depth: multiple safety measures with uncorrelated failure modes. This is akin to assembling multiple layers of Swiss cheese—each has holes, but when layered can compensate for each other's failures (Hendrycks et al., 2021). While this type of approach is promising, it also comes with problems. For example, many of the challenges in Section 3 are not unique to RLHF, so it may be hard to find safety methods with uncorrelated failures. In this section, we discuss approaches that can be used to better *understand* (Section 4.1), *improve* on (Section 4.2), and *complement* (Section 4.3) RLHF in various ways as part of a broader agenda for AI safety.

4.1 Frameworks for Better Understanding RLHF

Although RLHF is becoming more widely used, there remain open questions about what factors are at play within it and how they influence the overall outcome. Here, we discuss approaches to address challenges for RLHF.

Psychology and human-computer interaction. Many of the open questions with RLHF involve the dynamics at play between humans and AI. It remains a challenge to understand the conditions which best allow for safe, reliable human-computer interaction. Specifically, it is unclear what type of feedback (or combination thereof) is optimal for learning human goals, precisely how biases harm the quality of feedback, and how to best select and train human evaluators. As discussed in Section 3, human desires are difficult to express with a reward function (Skalse and Abate, 2022b; Bowling et al., 2023; Vamplew et al., 2022). Further work may be valuable toward inferring what beliefs humans are operating under and either asking for feedback while taking into account human uncertainty (Biyik et al., 2019) or correcting for human biases (Reddy et al., 2019; 2020; Chan et al., 2019; Tian et al., 2023). Reward modeling systems must also take advantage of techniques that distinguish between humans with different levels of expertise (Daniels-Koch and Freedman, 2022), confidence (Zhang et al., 2021), or noisiness (Barnett et al., 2023).

Sociology and social choice. AI alignment must address not only individuals' perspectives, but also the norms, expectations, and values of affected groups. Some works have begun to assess whether LLMs can be used to facilitate agreement between different humans (Bakker et al., 2022) and to codify the broad-ranging principles under which deployment of AI systems for public good can be assessed (Floridi and Cowls, 2022; Sartori and Theodorou, 2022). The majority-rule problem with RLHF can also be improved by algorithms that explicitly model multiple evaluators (Gordon et al., 2021; Davani et al., 2022; Daniels-Koch and Freedman, 2022; Gordon et al., 2022; Barnett et al., 2023), that tune models to individuals (Kumar et al., 2021), or that use more sophisticated aggregation strategies (Noothigattu et al., 2018). However, none of these approaches can solve the fundamental problem of how an AI system cannot be aligned to multiple groups of humans who hold conflicting viewpoints (Dobbe et al., 2021). Many societies, however, confront this fundamental issue regularly. For example, democracies seek to reflect social preferences by soliciting the feedback of individuals. These systems generally fail to align diverse preferences yet tend to be more acceptable than less-democratic alternatives. As such, it is important to analyze RLHF from the lens of social choice theory (Sen, 1986) and work to understand whether the means by which it aggregates preferences is normatively acceptable.